Number Theory: Diagnostic N

Time allowed: 45 minutes **Total:** 14 diagnostic marks

Instructions Answer both questions. Show full working and give complete proofs. No calculator, internet, AI assistance or hints during an independent diagnostic. Use separate answer paper if required.

Candidate: _____ Date: _____

Question 1 [7 marks]

NT-01

Determine all integer triples (x, y, z) satisfying

$$x^2 + y^2 = 3z^2.$$

Prove that no other integer solutions exist.

Question 2 [7 marks]

NT-02

For an integer $N \neq 0$, let $v_3(N)$ denote the largest nonnegative integer e for which 3^e divides N.

For every integer $k \geq 0$, determine

$$v_3\left(2^{3^k} + 1\right).$$

Prove your formula without quoting the lifting-the-exponent lemma.

End of paper. Write your answers on separate paper, labelled with the question number.